Sahrdaya College of Engineering and Technology, Kodakara			
Answer Key			
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY SEVENTH SEMESTER B.TECH DEGREE EXAMINATION, JANUARY 2023			
Course Code: EET455			
Course Name: ENERGY MANAGEMENT			
PART A			
		Answer all questions. Each question carries 3 marks	
	Q1	List the ethical and environmental factors associated with a nuclear power plant.	3 marks
	Ans	Nuclear power plants have many ethical and environmental factors associated with their operations. Here are some of the most significant ones: Ethical Factors: 1. Safety: Nuclear power plants carry the potential for accidents, which can have devastating consequences. Therefore, safety must be a top priority, and appropriate measures must be taken to ensure that workers and the surrounding communities are protected. 2. Health concerns: Nuclear power plants emit radiation, which can cause cancer and other health problems. Therefore, the ethical implications of exposing people to radiation must be considered. 3. Waste disposal: Nuclear power plants produce radioactive waste that remains dangerous for thousands of years. The ethical considerations of managing this waste responsibly and safely must be taken into account. 4. Environmental justice: The siting of nuclear power plants can disproportionately impact marginalized communities, such as low-income and minority populations. The ethical implications of such decisions must be carefully considered. Environmental Factors: 1. Water usage: Nuclear power plants require vast amounts of water to cool their reactors, which can impact local water resources and ecosystems. 2. Climate change: While nuclear power plants do not produce greenhouse gas emissions during operations, they do have a significant environmental impact during their construction and decommissioning phases. 3. Habitat destruction: The construction of nuclear power plants can require the clearing of large tracts of land, which can have a significant impact on local ecosystems and wildlife habitats. 4. Fuel mining: The extraction and processing of uranium for use in nuclear power plants can have significant environmental impacts, including soil and water contamination, and damage to local ecosystems.	
	Q2	Explain utility scale battery energy storage system.	3 marks

	Ans	A utility-scale battery energy storage system (BESS) is a type of energy storage system that is designed to store large amounts of electrical energy for use during times of peak demand or when renewable energy sources are not generating power. These systems typically consist of large lithium-ion batteries that are connected to the electrical grid and can store anywhere from a few megawatt-hours (MWh) to tens of MWh of energy. During periods of low demand or when renewable energy sources like wind or solar are generating excess power, the BESS charges its batteries by converting the electrical energy into chemical energy stored in the battery. Then, during times of high demand or when renewable energy sources are not generating enough power, the BESS discharges the stored energy back into the grid, providing additional power and reducing the need for fossil fuel-fired power plants. Utility-scale BESS systems can also provide grid stabilization and frequency regulation services by responding quickly to changes in demand or supply of electricity, thereby maintaining the grid's stability. In summary, utility-scale battery energy storage systems play a crucial role in balancing the electrical grid, integrating renewable energy sources, and providing reliable and cost-effective energy storage solutions.	
	Q3	Elucidate on the importance of transposition in transmission lines	3 marks
	Ans	Transposition is a technique used in electrical power transmission lines to improve the performance and reliability of the transmission system. Transposition refers to the process of changing the positions of the conductors in a transmission line to mitigate the effects of electromagnetic interference, also known as "stray capacitance." Stray capacitance refers to the capacitance that exists between adjacent conductors in a transmission line. This capacitance can cause power loss and reduce the efficiency of the transmission line by introducing unwanted electrical noise or "crosstalk" into the signal being transmitted. Transposition works by redistributing the capacitance and inductance of the transmission line. By rearranging the positions of the conductors, the capacitance and inductance are more evenly distributed along the length of the line, reducing the effects of stray capacitance and improving the overall performance of the transmission system. The benefits of transposition include: 1. Improved power transmission efficiency: Transposition reduces the losses that occur in the transmission line due to stray capacitance, resulting in more efficient power transmission. 2. Reduced electromagnetic interference: Transposition helps to mitigate the effects of electromagnetic interference on the transmission line, reducing the noise and crosstalk that can impact the signal being transmitted. 3. Increased system stability: Transposition can improve the stability of the transmission system by reducing the potential for voltage fluctuations and other electrical disturbances. 4. Improved reliability: Transposition can help to reduce the likelihood of electrical faults or failures in the transmission system, improving the reliability and safety of the power grid.	

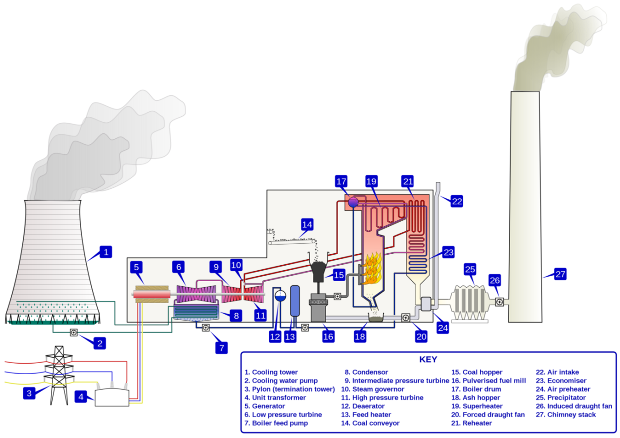
	Q4	Find the capacitance of a single-phase line 40km long consisting of two parallel conductors each 5mm in diameter and 1.5 m apart. The height of conductors above ground is 7m.The effect of ground may be included	3 marks
	Ans	To find the capacitance of the given transmission line, we can use the following formula: C = (2πεL) / ln(b/a) where C is the capacitance, ε is the permittivity of the medium (including the effect of ground), L is the length of the transmission line, a is the radius of the conductor, and b is the distance between the conductors. Given: ● Length of transmission line (L) = 40 km = 40,000 m ● Radius of each conductor (a) = 5 mm = 0.005 m ● Distance between conductors (b) = 1.5 m ● Height of conductors above ground = 7 m We can calculate the effective radius (aeff) of each conductor, taking into account the effect of ground, as follows: aeff = a * (1 + 0.5h / D) where h is the height of the conductor above ground, and D is the distance between the conductor and the ground. Substituting the given values, we get: aeff = 0.005 * (1 + 0.5*7 / 1.5) = 0.0073 m Now, we can calculate the capacitance using the formula: C = (2πεL) / ln(b/a) where ln is the natural logarithm. Substituting the given values and using the permittivity of free space (ε0) as the permittivity of the medium, we get: C = (2πε0εr L) / ln(b/a) where εr is the relative permittivity of the medium (including the effect of ground). The relative permittivity of the medium depends on the type of ground and the frequency of the transmission line. For example, for dry soil at 50 Hz, εr can be taken as around 4.6. Substituting this value, we get: C = (2πε0 * 4.6 * 40,000) / ln(1.5/0.0073) = 5.72 microfarads (approximately) Therefore, the capacitance of the single-phase line is approximately 5.72 microfarads.	
	Q5	What are the factors that affecting the phenomenon corona in transmission lines?	3 marks
	Ans	The phenomenon of corona occurs in high voltage transmission lines when the electric field strength around the conductor exceeds the breakdown strength of the surrounding air. This causes a partial discharge in the form of a corona or glow around the conductor. Corona can have various effects on the performance of transmission lines, including power loss, audible noise, and radio interference. The factors that affect the phenomenon of corona in transmission lines include: 1. Conductor diameter: The diameter of the conductor is an important factor affecting corona. As the diameter of the conductor increases, the electric field strength around the conductor decreases, reducing the likelihood of corona.	

		2. Conductor shape: The shape of the conductor can also affect the corona phenomenon. Sharp edges or corners on the conductor can create areas of high electric field strength, increasing the likelihood of corona. 3. Line voltage: The voltage level of the transmission line is a significant factor affecting the corona phenomenon. As the voltage level increases, the electric field strength around the conductor increases, making corona more likely. 4. Atmospheric conditions: Atmospheric conditions, such as humidity, temperature, and air pressure, can affect the corona phenomenon. Higher humidity levels can reduce the likelihood of corona by increasing the conductivity of the air. 5. Line design: The design of the transmission line, including the spacing between the conductors and the distance between the conductors and surrounding structures, can affect the corona phenomenon. Increasing the spacing between the conductors can reduce the electric field strength and the likelihood of corona. 6. Line loading: The amount of power being transmitted through the line can affect the corona phenomenon. Higher power levels increase the likelihood of corona by increasing the electric field strength around the conductor.	
	Q6	HVDC transmission system is preferred over HVAC transmission system for overhead transmission line lengths above 800 km. Justify the statement.	3 marks
	Ans	High Voltage Direct Current (HVDC) transmission system is preferred over High Voltage Alternating Current (HVAC) transmission system for overhead transmission line lengths above 800 km because of the following reasons: 1. Lower Transmission Losses: HVDC transmission systems experience lower power losses than HVAC transmission systems. The losses in an HVAC transmission system are proportional to the square of the transmission distance and the frequency of the alternating current used for power transmission. Therefore, for longer transmission distances, the power losses in an HVAC system can become very high. In contrast, the power losses in an HVDC system are almost independent of the transmission distance and are lower than those of an HVAC system for longer distances. 2. Higher Power Transfer Capability: HVDC transmission systems can transfer more power over longer distances as compared to HVAC systems. This is because HVDC systems can operate at higher voltages than HVAC systems, which allows them to transmit more power with less current. 3. Better Stability: HVDC transmission systems are less susceptible to voltage instability and stability problems than HVAC transmission systems. This is because HVDC systems have a more controllable power flow, which reduces the likelihood of voltage instability and can help maintain a stable voltage profile across the system. 4. Smaller Footprint: HVDC transmission systems require fewer towers and cables than HVAC transmission systems, which makes them more economical and reduces the environmental impact. 5. Interconnectivity: HVDC transmission systems can easily interconnect different power systems with different frequencies, voltages, and phase angles. This allows HVDC transmission systems to integrate power from different sources, including renewable energy sources, which is essential for the future of the power industry.	

	Q7	Define the terms Restriking voltage, Recovery voltage and rate of rise of restriking voltage with reference to operation of circuit breakers	3 marks
	Ans	Restriking voltage, recovery voltage, and rate of rise of restriking voltage are important terms associated with the operation of circuit breakers. They are defined as follows: 1. Restriking voltage: Restriking voltage is the voltage that appears across the contacts of a circuit breaker after current interruption. It is caused by the re-ignition of the arc between the contacts due to the presence of charged particles that were not completely cleared during the initial interruption. The restriking voltage can be higher than the system voltage, and if not dealt with, can lead to re-ignition of the arc and damage to the circuit breaker. 2. Recovery voltage: Recovery voltage is the voltage that appears across the contacts of a circuit breaker during the period between the end of arc interruption and the complete extinction of the arc. It is caused by the energy stored in the circuit due to the presence of inductive or capacitive elements in the circuit. The recovery voltage can be higher than the system voltage, and if not dealt with, can lead to re-ignition of the arc. 3. Rate of rise of restriking voltage (RRRV): RRRV is the rate at which the restriking voltage rises after interruption of the current. It is caused by the rapid recovery of the voltage across the inductive or capacitive elements in the circuit. The RRRV can be very high, and if not dealt with, can lead to re-ignition of the arc and damage to the circuit breaker.	
	Q8	List the causes of over voltages in a power system	3 marks
	Ans	Overvoltages are a significant concern in power systems, and they can damage electrical equipment and cause power outages. Some common causes of overvoltages in a power system include: 1. Lightning strikes: Lightning strikes near power lines can cause high-voltage surges that can damage electrical equipment and cause power outages. 2. Switching operations: Switching operations, such as opening or closing of a circuit breaker, can cause overvoltages due to the sudden change in current flow. This can lead to voltage transients or high-frequency oscillations. 3. Faults in the power system: Faults such as short circuits or ground faults can cause overvoltages due to the sudden change in current flow. This can lead to voltage transients or high-frequency oscillations. 4. Load rejection: Load rejection, where a large load is suddenly disconnected from the power system, can cause overvoltages due to the sudden change in power flow. 5. Capacitor switching: Switching of large capacitors in a power system can cause overvoltages due to the sudden change in reactive power flow. 6. Ferroresonance: Ferroresonance is a phenomenon that can occur when the system has nonlinear inductive elements and capacitors. It can lead to overvoltages, which can damage electrical equipment. 7. Electrostatic coupling: Electrostatic coupling can occur due to the proximity of power lines to other conductive objects, such as communication cables or metal structures. This can lead to overvoltages that can damage electrical equipment.	
	Q9	Distinguish between radial and ring main distribution systems	3 marks

	Ans	**Radial Distribution System** This system is used only when substation or generating station is located at the center of the consumers. In this system, different feeders radiate from a substation or a generating station and feed the distributors at one end. Thus, the main **characteristic of a radial distribution system** is that the power flow is in only one direction. Single line diagram of a typical radial distribution system is as shown in the figure below. It is the simplest system and has the lowest initial cost. **Ring Main Distribution System** A similar level of system reliability to that of the parallel feeders can be achieved by using **ring distribution system**. Here, each distribution transformer is fed with two feeders but in different paths. The feeders in this system form a loop which starts from the substation bus-bars, runs through the load area feeding distribution transformers and returns to the substation bus-bars.	
	Q10	Elucidate the use of ABC cable in power system	3 marks
	Ans	ABC cable, which stands for Aerial Bundle Cable, is a type of power cable used in power distribution systems. It is designed to be installed above ground, typically on utility poles or other support structures. ABC cables have become increasingly popular in recent years, as they offer several advantages over traditional bare conductor systems. The main advantage of ABC cables is that they are safer and more reliable than bare conductor systems. Traditional bare conductor systems use uninsulated conductors, which can pose a safety hazard to people and wildlife. ABC cables, on the other hand, use insulated conductors that are bundled together with a support messenger wire. This design provides several benefits, including: 1. Improved safety: The insulation on the conductors helps to prevent accidental contact with live conductors, reducing the risk of electrical shock or electrocution. 2. Reduced power outages: The bundled design of ABC cables makes them more resistant to damage from high winds, ice, or other weather-related events. This reduces the risk of power outages due to downed power lines. 3. Lower maintenance costs: Because ABC cables are more resistant to damage, they require less maintenance than traditional bare conductor systems. This can lead to lower maintenance costs for utility companies. 4. Improved aesthetics: The bundled design of ABC cables is more visually appealing than traditional bare conductor systems, which can be an important consideration in residential areas or tourist destinations.	
PART B			
Answer any one full question from each module. Each question carries 14 marks			
Module 1			
	Q11 (a)	With a neat block diagram explain the working of a thermal power plant	9 marks

Ans



Thermal power plant

A Thermal power plant is an electric producing plant. Certain thermal power stations are also designed to produce heat for industrial purposes, for district heating, or desalination of water, in addition to generating electrical power. Here are thermal power plant components and working principles.

- **River or Canal**
- **Heater**
- **Boiler**
- **Superheater**
- **Economizer**
- **Air pre-heater**
- **Turbine**
- **Condenser**
- **Cooling towers and ponds**
- **Alternator**
- **Feedwater pump**

Here are the main thermal power plant parts and functions.

River or Canal

As we know here a huge amount of water is present, and it is further used for th**Heater**

As the name indicates, a low or high-pressure heater means that it increases or decreases the pressure of the water.

Boiler

In the boiler, there are two sections: one section refers to coal storage and coal handling that store the coal and further used it when needed. The other section is the ash handling and ash storage plant that the produced ash of the process of coal-burning goes to ash storage.

The mixture of pulverized coal and air is taken into the boiler and then burnt in the combustion zone. On ignition of the fuel, a large fireball is formed at the center of the boiler and a large amount of heat energy is radiated from it. The heat energy finds use to convert the water into steam at high temperature and pressure. Steel tubes run along the boiler walls in which water is converted into steam. The flue gases from the boiler make their way through the superheater, economizer, air preheater, and finally, get exhausted to the atmosphere from the chimney.

Superheater

		The superheater tubes are hung at the hottest part of the boiler. The saturated steam produced in the boiler tubes is superheated to about 540 °C in the superheater. The superheated high-pressure steam is then fed to the steam turbine. **Economizer** An economizer is essentially a feedwater heater that heats the water before supplying it to the boiler. When water pressure gets increased there some amount of heat generates and that heat sends from the economizer to the boiler **Air pre-heater** The primary air fan takes air from the atmosphere, and it is then warmed in the air pre-heater. Pre-heated air is injected with coal in the boiler. The advantage of pre-heating the air is that it improves coal combustion. **Turbine** The main function of the turbine is that when steam strikes the turbine the blade rotates and it converts the heat energy into mechanical energy. High-pressure superheated steam is fed to the steam turbine which causes turbine blades to rotate. The energy in the steam is converted into mechanical energy in the steam turbine which acts as the prime mover. The pressure and temperature of the steam fall to a lower value and it expands in volume as it passes through the turbine. The expanded low-pressure steam is exhausted in the condenser. **Condenser** The condenser presents here to cool the working fluid or we can say to remove the heat from the water. The exhausted steam is condensed in the condenser using cold water circulation. Here, the steam loses its pressure as well as temperature and it is converted back into the water. Condensing is essential because compressing a fluid that is in a gaseous state requires a huge amount of energy concerning the energy required in compressing liquid. Thus, condensing increases the efficiency of the cycle. **Cooling towers and ponds** A condenser needs a huge quantity of water to condense the steam. Most plants use a cooled cooling system where warm water coming from the condenser is cooled and reused. A cooling tower is a steel or concrete **Alternator** The steam turbine is coupled to an alternator. When the turbine rotates the alternator, electrical energy is generated. This generated electrical voltage is then stepped up with the help of a transformer and then transmitted where it is to be utilized. **Feedwater pump** The condensed water is again fed to the boiler by a feedwater pump. Some water may be lost during the cycle, which is suitably supplied from an external water source.	
	Q11 (b)	A diesel station supplies the following loads to various consumers : Industrial consumer = 1500 kW ; Commercial establishment = 750 kW Domestic power = 100 kW; Domestic light = 450 kW. If the maximum demand on the station is 2500 kW and the number of kWh generated per year is 45 × 106 , determine (i) the diversity factor and (ii) annual load factor.	5 marks
	Ans	Maximum demand = 2500 kW Annual energy generated = 45 × 10^6 kWh Industrial consumer = 1500 kW	

		Commercial establishment = 750 kW Domestic power = 100 kW Domestic light = 450 kW (i) The diversity factor can be calculated as: Diversity factor = (Maximum demand of all consumers) / (Overall maximum demand) Maximum demand of all consumers = 1500 + 750 + 100 + 450 = 2800 kW Overall maximum demand = 2500 kW Diversity factor = 2800 / 2500 = 1.12 (ii) The annual load factor can be calculated as: Annual load factor = (Total energy consumed in a year) / (Maximum demand x Number of hours in a year) Number of hours in a year = 365 days x 24 hours per day = 8760 hours Total energy consumed in a year = 45 × 10^6 kWh Maximum demand = 2500 kW Annual load factor = (45 × 10^6 kWh) / (2500 kW x 8760 h/year) Annual load factor = 2.04 Therefore, the diversity factor is 1.12 and the annual load factor is 2.04.	
	Q12 (a)	Differentiate between rooftop and ground mounted solar power plants	14 marks
	Ans	Rooftop solar power plants and ground-mounted solar power plants are two different types of solar power systems. The main difference between them is the location where the solar panels are installed. Rooftop solar power plants are installed on the rooftops of buildings, such as homes, offices, and commercial buildings. These solar panels are usually installed on the south-facing side of the roof, where they can receive maximum sunlight throughout the day. Rooftop solar power plants can be installed on both residential and commercial buildings, and they can range in size from a few kilowatts to several megawatts.	

		Ground-mounted solar power plants, on the other hand, are installed on the ground, usually in large open spaces or fields. These solar panels are usually installed on a fixed tilt rack, or a tracking system that follows the sun throughout the day. Ground-mounted solar power plants are usually larger than rooftop solar power plants and can range in size from a few megawatts to hundreds of megawatts. Some of the key differences between rooftop and ground-mounted solar power plants are: 1. Location: Rooftop solar power plants are installed on the rooftops of buildings, while ground-mounted solar power plants are installed on the ground. 2. Size: Rooftop solar power plants are usually smaller than ground-mounted solar power plants. 3. Installation: Rooftop solar power plants require a specialized installation process that takes into account the design of the building and the weight of the solar panels. Ground-mounted solar power plants are usually easier to install, as they can be installed on a level surface. 4. Efficiency: Ground-mounted solar power plants are usually more efficient than rooftop solar power plants, as they can be optimized for maximum sun exposure throughout the day. 5. Cost: The cost of rooftop solar power plants is usually lower than ground-mounted solar power plants, as they require less material and labor for installation.	
	Q12 (b)	A generating station has the following daily load cycle : Time (Hours) 0—6 6—10 10—12 12—16 16—20 20—24 Load (MW) 40 50 60 50 70 40 Draw the load curve and find (i) maximum demand (ii) units generated per day (iii) average load and (iv) load factor	
	Ans	To draw the load curve, we plot the load values against the corresponding time intervals.From the load curve, we can see that the maximum demand is 70 MW, which occurs between 16:00 and 20:00 hours. To find the units generated per day, we need to calculate the area under the load curve in terms of MW-hours. We can do this by dividing the day into small intervals (e.g. 1 hour), calculating the load at each interval, and then multiplying it by the duration of the interval. We then add up all these values to get the total number of MW-hours generated in a day. Units generated per day = Area under the load curve = (40 x 6) + (50 x 4) + (60 x 2) + (50 x 4) + (70 x 4) + (40 x 4) = 960 MW-hours The average load is given by the total energy generated per day divided by the number of hours in a day: Average load = Total energy generated / Number of hours = 960 / 24 = 40 MW	

		The load factor is the ratio of the average load to the maximum demand: Load factor = Average load / Maximum demand = 40 / 70 = 0.57 or 57%	
Q13 (a)		Derive the expression for capacitance of a three phase overhead transmission line assuming symmetrical spacing between conductors	9 marks
Ans		Consider a three-phase overhead transmission line with symmetrical spacing between conductors, Let the distance between the conductors be d and the height of the conductors above ground be h. We can calculate the capacitance of the line using the following steps: 1. First, we calculate the capacitance between each pair of conductors. For a pair of parallel conductors, the capacitance per unit length is given by: C' = ε * A / d where ε is the permittivity of free space, A is the area of each conductor, and d is the distance between the conductors. 2. Next, we calculate the capacitance between each conductor and the ground. For a single conductor above a flat ground plane, the capacitance per unit length is given by: C'' = ε * π * d / ln(2h/d) where ln is the natural logarithm. 3. The total capacitance per unit length between each phase conductor and ground is the sum of the capacitances between the conductor and the other two phases, plus the capacitance between the conductor and the ground: C = 2C' + C'' where the factor of 2 comes from the fact that there are two other phase conductors that contribute capacitance to each conductor. 4. Finally, we can calculate the capacitance per unit length between any two phases by adding up the capacitances per unit length between each phase conductor and ground: C_phase-phase = C + C' where C' is the capacitance per unit length between the two phase conductors. Substituting the expression for C' into the above equation, we get: C_phase-phase = ε * (2A/d + πd/ln(2h/d)) This is the expression for the capacitance per unit length between any two phases of a three-phase overhead transmission line with symmetrical spacing between conductors.	
Q13 (b)		Figure shows the spacings of a double circuit 3-phase overhead line. The phase sequence is ABC and the line is completely transposed. The conductor radius in 1·3	5 marks

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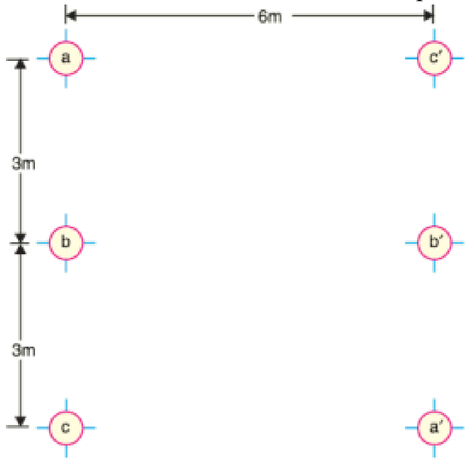
<tr><td></td><td></td><td>cm. Find the inductance per phase per kilometre.

</td><td></td></tr>
<tr><td></td><td>Ans</td><td>Given data:
Radius of each conductor (r) = 1.3 cm = 0.013 m
Phase sequence = ABC
Spacing of conductors:
Horizontal distance between A and B = 3.2 m
Horizontal distance between B and C = 5 m
Vertical distance between A-B and B-C = 6.5 m

To find: Inductance per phase per kilometer

Let's first calculate the distance between each pair of conductors.

For Phase A:
Distance between A and B = sqrt[(3.2/2)^2 + 6.5^2] = 7.258 m

Distance between A and C = sqrt[(3.2/2 + 5)^2 + 6.5^2] = 10.516 m

For Phase B:

Distance between B and C = sqrt[5^2 + 6.5^2] = 8.154 m

The inductance of the line is given by the expression:

L = 2μ0/π * ln(D/d)

where, μ0 is the permeability of free space (4π × 10^-7 H/m), D is the distance between conductors and d is the radius of each conductor.

For Phase A:</td><td></td></tr>
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		L_AB = L_AC = 2μ0/π * ln(7.258/0.026) For Phase B: L_BC = 2μ0/π * ln(8.154/0.026) Total inductance per phase per kilometer can be calculated as: L_phase = (L_AB + L_AC + L_BC) / 1000 Here, the factor of 2 comes from the fact that the line is completely transposed, and we have to consider both the self-inductance and mutual inductance of each phase. Substituting the values of μ0, L_AB, L_AC, and L_BC, we get: L_phase = 0.88 μH/km (approximately) Therefore, the inductance per phase per kilometer of the given double circuit 3-phase overhead line is 0.88 μH/km.	
	Q14 (a)	Derive the expression for sending end voltage and sending end current in terms of receiving end voltage and receiving end current for long transmission line	5 marks
	Ans	When a three-phase transmission line of length 'l' and characteristic impedance 'Z' is energized by a sending end voltage 'VS' and a sending end current 'IS', the receiving end voltage 'VR' and receiving end current 'IR' are given by: VR = VS - IZ and IR = (VS / Z) - I Here, 'I' is the line current, which is same at both sending and receiving end. To derive the expressions for sending end voltage and current in terms of receiving end voltage and current, we can use the above equations and solve for 'VS' and 'IS'. From VR = VS - IZ, we can write: VS = VR + IZ From IR = (VS / Z) - I, we get: VS = ZIR + I^2Z Substituting the value of 'VS' from the first equation into the second equation, we get: VR + IZ = ZIR + I^2Z Simplifying this equation, we get:	

		$I = (VR / Z) + (IZ / Z) - I^2$ Rearranging this equation, we get a quadratic equation in 'I': $I^2 - (VR / Z)I - (IS / Z) = 0$ Using the quadratic formula to solve for 'I', we get: $I = (VR / 2Z) \pm \mathrm{sqrt}[(VR^2 / 4Z^2) + (IS / Z)]$ The positive sign is used for the case when the line is inductive, and the negative sign is used for the case when the line is capacitive. Substituting this value of 'I' into the equations for 'VS' and 'IS', we get: $VS = VR + IZ$ and $IS = (VR - VS) / Z$ Substituting the value of 'I' in the above equations, we get: $VS = VR + (VR / 2Z) \pm \mathrm{sqrt}[(VR^2 / 4Z^2) + (IS / Z)] * Z$ and $IS = (VR - VS) / Z$ Simplifying the expression for 'VS', we get: $VS = VR (1 + jX) \pm jVR\ \mathrm{sqrt}[(VR^2 / 4Z^2) + (IS / Z)]$ where X is the reactance of the line per unit length, given by $X = \omega l/C$, where ω is the angular frequency and C is the capacitance per unit length. In the case of a long transmission line, the capacitance of the line dominates over its inductance, so the second term under the square root can be neglected, and the expression for 'VS' becomes: $VS \approx VR (1 + jX)$ Similarly, the expression for 'IS' can be simplified as: $IS \approx VR / Z*(1 - jX)$	

		Thus, for a long transmission line, the expressions for sending end voltage and current in terms of receiving end voltage and current are given by: VS ≈ VR (1 + jX) IS ≈ VR / Z*(1 - jX)	
	Q14 (b)	A 3-phase, 50-Hz overhead transmission line 100 km long has the following constants : Resistance/km/phase = 0.1 Ω Inductive reactance/km/phase = 0·2 Ω Capacitive susceptance/km/phase = 0·04 × 10− 4 siemen Determine (i) the sending end current (ii) sending end voltage (iii) sending end power factor and (iv) transmission efficiency when supplying a balanced load of 10,000 kW at 66 kV, p.f. 0·8 lagging. Use nominal T method.	9 marks
	Ans	Given parameters are: Length of transmission line, L = 100 km Resistance/km/phase, R = 0.1 Ω Inductive reactance/km/phase, X = 0.2 Ω Capacitive susceptance/km/phase, B = 0.04 × 10− 4 siemen Frequency, f = 50 Hz Load, P = 10,000 kW Voltage, V = 66 kV Power factor, pf = 0.8 lagging (i) Sending end current: The current in each phase of the transmission line can be calculated using the formula: I = (P/√3Vpf) Where, √3 is the factor for 3-phase system. Substituting the given values, we get: I = (10,000/√3×66×0.8) = 79.99 A Therefore, the sending end current is 79.99 A. (ii) Sending end voltage: The sending end voltage can be calculated using the voltage regulation formula: Vs = Vr + IZeq	

Where, Vs is the sending end voltage, Vr is the receiving end voltage, I is the current in each phase, and Zeq is the equivalent impedance of the transmission line.

To calculate Zeq, we can use the pi-equivalent circuit:

$Zeq = [(R+jX)+(jB/2)]//2$

Where, // denotes the parallel combination of two impedances.

Substituting the given values, we get:

$Zeq = [(0.1+j0.2)+(j0.04\times10^{-4}/2)]//2$
$= (0.1+j0.2+j0.02\times10^{-4})/2$

$= 0.05+j0.1+j0.01\times10^{-4}$

Substituting Zeq and other given values in the voltage regulation formula, we get:

$Vs = Vr + IZeq$
$= 66\ kV + 79.99\times(0.05+j0.1+j0.01\times10^{-4})$

$= 66.4\ kV$ (approx)

Therefore, the sending end voltage is 66.4 kV (approx).

(iii) Sending end power factor:

The sending end power factor can be calculated using the formula:

$pf = P/(\sqrt{3}VsIs)$

Substituting the given values, we get:

$pf = 10{,}000/(\sqrt{3}\times66.4\times79.99)$

$= 0.8002$ (approx)

Therefore, the sending end power factor is 0.8002 (approx).

(iv) Transmission efficiency:

The transmission efficiency can be calculated using the formula:

$\eta = P/(P+3I^2R)$

Substituting the given values, we get:

$\eta = 10{,}000/(10{,}000+3\times(79.99)^2\times0.1\times100)$

		= 0.9765 (approx) Therefore, the transmission efficiency is 0.9765 (approx). Hence, the sending end current is 79.99 A, sending end voltage is 66.4 kV (approx), sending end power factor is 0.8002 (approx), and transmission efficiency is 0.9765 (approx)	
	Q15 (a)	The capacitances of a 3-phase belted cable are 12·6 μF between the three cores bunched together and the lead sheath and 7·4 μF between one core and the other two connected to sheath. Find the charging current drawn by the cable when connected to 66 kV, 50 Hz supply.	8 marks
	Ans	To calculate the total capacitance, we can use the following formula: C = C1 + C2 where C1 is the capacitance between the three cores and the lead sheath, and C2 is the capacitance between one core and the other two connected to the sheath. Substituting the given values, we get: C = 12.6 μF + 7.4 μF = 20 μF Now, we can calculate the charging current using the formula: I = 2πfCV where f is the frequency, C is the capacitance, and V is the voltage. Substituting the given values, we get: I = 2π×50×20×10^-6×66×10^3 = 41.48 A (approx) Therefore, the charging current drawn by the cable when connected to a 66 kV, 50 Hz supply is 41.48 A (approx).	
	Q15 (b)	Explain the methods of improving string efficiency	6 marks
	Ans	1. Increasing the voltage level: As the voltage level increases, the current required to deliver the same amount of power decreases, resulting in lower I2R losses. This is why high voltage transmission lines are preferred for long-distance power transmission.	

		2. Using larger conductor size: The resistance of the conductor is directly proportional to its length and inversely proportional to its cross-sectional area. Using a larger conductor size reduces the resistance and hence, the I2R losses. 3. Using bundled conductors: In bundled conductors, two or more conductors are twisted together to form a bundle. This reduces the inductance and capacitance of the line, resulting in lower power losses. 4. Using transformers: Transformers can be used to step up the voltage at the sending end and step it down at the receiving end. This reduces the current required to deliver the same amount of power and hence, the I2R losses. 5. Using reactive compensation: Inductive and capacitive reactance in the transmission line can be compensated by using series and shunt capacitors and inductors. This reduces the voltage drop in the line and hence, the power losses. 6. Using power electronics: Power electronics devices such as FACTS (Flexible AC Transmission Systems) and HVDC (High Voltage Direct Current) can be used to improve the power transfer capability and reduce the power losses in the transmission line.	
	Q16 (a)	An overhead transmission line conductor having a parabolic configuration weighs 1·925 kg per metre of length. The area of X-section of the conductor is 2·2 cm2 and the ultimate strength is 8000 kg/cm2 . The supports are 600 m apart having 15 m difference of levels. Calculate the sag from the taller of the two supports which must be allowed so that the factor of safety shall be 5. Assume that ice load is 1 kg per metre run and there is no wind pressure.	9 marks
	Ans	To calculate the sag of the overhead transmission line conductor, we can use the following formula: Sag = [(Tension × L^2) / (8 × w × factor of safety)] + [(L / 2)^2] where Tension is the tension in the conductor, L is the span length (600 m), w is the weight per unit length of the conductor, and factor of safety is the safety factor (5 in this case). First, we need to calculate the tension in the conductor. The tension in the conductor is given by the formula: Tension = (Weight of the conductor + Weight of the ice) / 2 where Weight of the conductor is given by: Weight of the conductor = Length of the conductor × Weight per unit length Substituting the given values, we get: Weight of the conductor = 1 m × 2.2 cm^2 × (1.925 kg/m) = 0.04195 kg/m	

		Length of the conductor = $\sqrt{L^2 + H^2} = \sqrt{600^2 + 15^2}$ = 600.15 m (approx) Weight of the ice = 1 kg/m × L = 600 kg Substituting these values, we get: Tension = (0.04195 kg/m + 1 kg/m) / 2 × 9.81 m/s^2 = 0.0519 kN (approx) Now, we can substitute the values in the formula for sag: Sag = [(Tension × L^2) / (8 × w × factor of safety)] + [(L / 2)^2] = [(0.0519 × 600^2) / (8 × 1.925 × 5)] + [(600 / 2)^2] = 21.42 m (approx) Therefore, the sag from the taller of the two supports which must be allowed so that the factor of safety shall be 5 is 21.42 m (approx).	
	Q16 (b)	Explain the significance of Flexible AC Transmission System (FACTS) devices in power system. Explain the working of any two FACTS devices. List out the demerits of FACTS devices	5 marks
	Ans	Flexible AC Transmission System (FACTS) devices are used to control and enhance the power transfer capability of transmission systems. These devices use power electronics technology to provide fast and efficient control of the power system parameters, such as voltage, current, and impedance. The significance of FACTS devices in power systems can be summarized as follows: 1. Improved power transfer capability: FACTS devices can control the power flow in the transmission lines and improve the power transfer capability of the system. This reduces the need for new transmission lines and helps to utilize the existing system more efficiently. 2. Voltage and stability control: FACTS devices can regulate the voltage and maintain the stability of the power system, especially during disturbances or faults. This improves the reliability and security of the power system. 3. Lower system losses: FACTS devices can reduce the transmission losses by optimizing the power flow in the system. This leads to a reduction in the operating cost and improves the economic viability of the power system. Working of two FACTS devices: 1. Static VAR Compensator (SVC): SVC is a shunt-connected FACTS device that is used for reactive power compensation. It consists of a thyristor-controlled reactor and a thyristor-switched capacitor bank. The thyristor-controlled reactor provides continuous and smooth control of the reactive power, while the thyristor-switched capacitor bank provides	

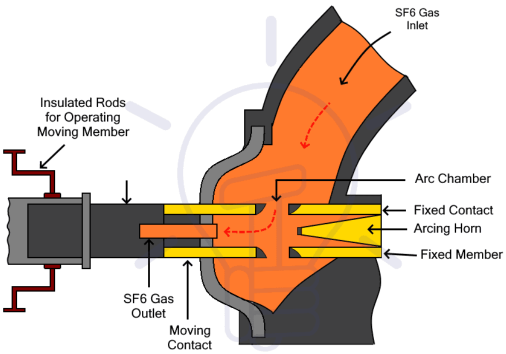
		discrete control. During normal operation, the SVC is adjusted to maintain the required voltage level and power factor at the point of connection. When there is a sudden change in the load or system conditions, the SVC responds quickly to provide the required reactive power and maintain the system stability. The thyristor-controlled reactor and thyristor-switched capacitor bank are controlled by a control system that monitors the voltage and current at the point of connection and adjusts the reactive power accordingly. 2. Unified Power Flow Controller (UPFC): UPFC is a series-connected FACTS device that is used for power flow control. It consists of a combination of a series transformer, a shunt reactor, and a shunt capacitor. The series transformer provides continuous control of the impedance, while the shunt reactor and capacitor provide discrete control of the reactive power. During normal operation, the UPFC is adjusted to maintain the required power flow and voltage level at the point of connection. When there is a congestion in the transmission line or a sudden change in the load or system conditions, the UPFC responds quickly to provide the required power flow and maintain the system stability. The impedance and reactive power are controlled by a control system that monitors the voltage and current at the point of connection and adjusts the parameters accordingly. Demerits of FACTS devices: 1. Cost: FACTS devices are expensive to install and maintain, and the cost may not always be justified by the benefits obtained. 2. Complexity: FACTS devices involve complex power electronics technology and control systems, which require specialized expertise to design, operate, and maintain. 3. Harmonic distortion: FACTS devices may introduce harmonic distortion in the power system, which can affect the performance of other devices and cause interference with communication systems. 4. Reliability: FACTS devices require a high level of reliability to maintain the power system stability and security, and any malfunction can have serious consequences.	
	Q17 (a)	Explain the term duality in terms of amplitude and phase comparators in static relays	8 marks
	Ans	Duality in static relays refers to the relationship between amplitude and phase comparators used in the relay circuit. In general, an amplitude comparator compares the magnitude of the measured input signal to a reference value, while a phase comparator compares the phase angle of the input signal to a reference angle. In duality, the amplitude and phase comparators are interchanged, and the signal inputs are transformed accordingly. This means that if the original circuit has an	

<table>
<tr><td></td><td></td><td>amplitude comparator followed by a phase comparator, the dual circuit will have a phase comparator followed by an amplitude comparator.

The duality concept is based on the fact that the behavior of an electric circuit remains unchanged if the current and voltage signals are interchanged and the circuit elements are replaced with their duals. In other words, the dual of an amplitude comparator is a phase comparator and vice versa.

The application of duality in static relays is useful for analyzing the response of the relay circuit to different types of faults. By interchanging the amplitude and phase comparators, the relay designer can examine how the circuit behaves under different fault conditions and determine the optimal arrangement of the comparators for specific applications.</td><td></td></tr>
<tr><td></td><td>Q17 (b)</td><td>With the help of a neat diagram explain the working of SF6 circuit breaker. List the advantages of SF6 when used in Gas Insulated substation</td><td>6 marks</td></tr>
<tr><td></td><td>Ans</td><td>What is SF6 Circuit Breaker?

Sulphur Hexafluoride or SF6 circuit breaker is a type of circuit breaker that uses pressurized SF6 gas to extinguish the arc. It is a dielectric gas having superior insulating and arc quenching properties far better than air or oil. It is used for arc quenching in high voltage circuit breakers up to 800 kV in power stations, electrical grids etc.



Sulphur Hexafluoride SF6 Circuit Breaker

SF6 gas has very high electronegativity. It has a strong tendency to absorb free electrons. When an arc is struck between the contacts, it absorbs the free electrons from it. It converts into negative ions which are heavier than electrons. Due to its heavyweight, its mobility is reduced. Therefore, the mobility of the charges in SF6 gas has low mobility which enhances the dielectric strength of the medium since the movement of charges is responsible for current flow.

Physical properties of SF6 gas

● It is non-flammable gas.
● It is colorless and odorless gas.
● It has excellent thermal conductivity.
● It has high density and heavier than air.</td><td></td></tr>
</table>

- It liquefies at low temperature which is pressure-dependent.

Chemical Properties

Chemical properties of SF6 gas

- SF6 gas is stable and inert.
- It is non-toxic in its pure form but its products are.
- It has high electronegativity meaning it has a strong affinity for free electrons.
- It recombines very easily after arc quenching for re-utilization.
- They are non-corrosive.

Electrical Properties

- It has superior dielectric strength which is directly proportional to the pressure.
- Its arc quenching capabilities is almost 100 times better than air.
- The frequency of the voltage does not affect its dielectric strength.

Construction of SF6 Circuit Breaker

The SF6 circuit breakers consist of two main parts

- Interrupter Unit
- Gas System

Interrupter Unit

The interrupter unit consists of two types of current-carrying contacts i.e. the fixed and movable contact. The fixed contacts as its name suggest do not move while the movable contact moves back and forth using an arm actuated by a mechanism.

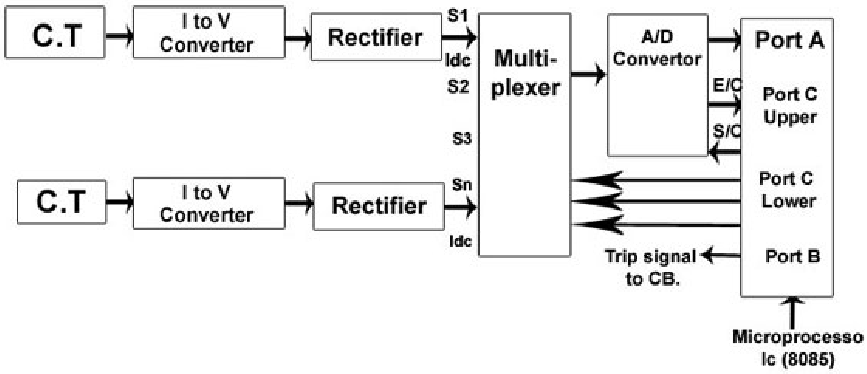
There is a vent for the inlet and outlet of the pressurized SF6 gas to cool off the arc as well as extinguish it.

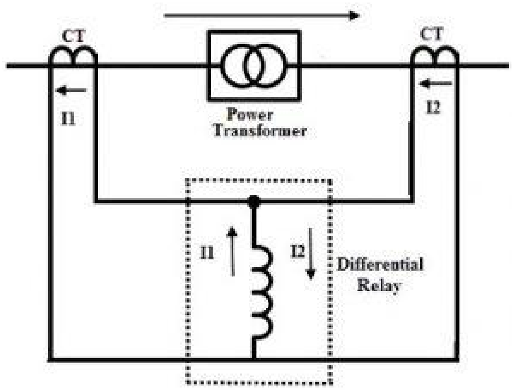
Gas System

The SF6 is a very expensive gas and the emission of its product gases is hazardous for the environment. Therefore, a closed gas system is used, where the used SF6 is recombined for reuse. Its pressure is also maintained as its dielectric strength greatly depends upon it.

Working Principle of SF6 Circuit Breaker

The SF6 gas is compressed and stored inside a tank. During the fault conditions, the contacts are separated and an arc is struck between them. The highly pressurized SF6 gas is also released at the same moment.

		The arc which is the movement of charges contains free electrons. The SF6 being highly electronegative, absorbs the free electrons forming negative ions. These ions are heavier and have low mobility as compared to free electrons. SF6 + e– = SF6– SF6 + e– = SF5– + F Due to the heavy mass of the formed negative ions, the movement of the charges reduces between the contacts. This increases the dielectric strength of the medium that quenches the arc at zero current. The blast of SF6 gas also reduces the temperature of the arc which also reduces its strength. The pressure of the gas is also directly proportional to the dielectric strength of the SF6 gas.	
	Q18 (a)	Draw and explain the block diagram of microprocessor based over current relay	8 marks
	Ans	 ○ Current is taken from C.T. and given to I to V converter because many electronics circuit require voltage signal for operation. ○ The A.C. voltage is converted into D.C. voltage by using rectifier. ○ This D.C. voltage is proportional to load current only. ○ The output of rectifier is given to Multiplexer. ○ The Multiplexer gives output to A/D Converter where Analog DC voltage is converted to Digital form (in form of O and 1 i.e. binary form). ○ Microprocessor understands only codes in 0 and 1 form. ○ HP (Microprocessor) gives S/C (start of conversion) signal to A/D converter (I.e. analog to digital conversion is started and HP gives permission to A/D convertor for this by sending S/C) ○ When converting from analog to digital is over (finish) then A/D converter sends E/C signal to HP (E/C – End of Conversion). ○ When work of A/D is over then up compare the magnitude of this incoming current with required current value (I.e. set value or reference value). If incoming value is more – fault is occurring and trip signal is send to CB circuit	

		breaker.	
	Q18 (b)	Explain with a neat circuit diagram the working principle of differential relay.	6 marks
	Ans	**Differential Relay Working Principle** Differential relay works on the principle of comparison between the phase angle & two or more same electrical quantities magnitudes. Comparing these two electrical quantities within a circuit with a differential relay is very simple in application & positive in action. For instance, in comparison to the entering current & leaving current in a line, if a huge current goes through the protected line as compared to the current leaving from it, then additional current must supply within the fault. So, the difference between the two electrical quantities can control a relay to separate the circuit. In normal operating conditions, the entering & leaving currents are equivalent in phase & magnitude thus the relay will not work. However, if any fault occurs within the system, then these flow of currents will be no longer equivalent in phase & magnitude. This kind of relay is used in such a way that the difference between the entering & leaving current supplies throughout the relay's operating coils. So, the relay coil can be energized in fault conditions because of the various quantity of the current. So, this relay functions & opens the circuit breaker for tripping the circuit.  ***Differential Relay Circuit*** In the above **differential relay circuit**, there are two current transformers which are connected to any face of the power transformer like one CT is connected on the primary side and the other is connected at the secondary side of the PT (power transformer). This relay simply compares the flow of currents on both sides. If there is any unbalance in the current flow of the circuit then this relay tends to operate. These relays can be current differential, voltage balance & biased differential relays.	

		Types of Differential Relay These relays are classified into three types current differential, voltage balance, and percentage differential relay or biased beam relay.	
	Q19 (a)	A single phase a.c. generator supplies the following loads : (i) Lighting load of 20 kW at unity power factor. (ii) Induction motor load of 100 kW at p.f. 0.707 lagging. (iii) Synchronous motor load of 50 kW at p.f. 0.9 leading. Calculate the total kW and kVA delivered by the generator and the power factor at which it works	6 marks
	Ans	(i) Lighting load of 20 kW at unity power factor has no reactive power, so the total power is simply 20 kW. (ii) Induction motor load of 100 kW at p.f. 0.707 lagging. Using the power triangle, we can find that the reactive power is Q = P * tan(acos(0.707)) = 70.7 kVAR, and the apparent power is S = P / cos(acos(0.707)) = 141.4 kVA. (iii) Synchronous motor load of 50 kW at p.f. 0.9 leading. Using the power triangle, we can find that the reactive power is Q = P * tan(acos(0.9)) = 28.87 kVAR, and the apparent power is S = P / cos(acos(0.9)) = 55.56 kVA. Therefore, the total real power delivered by the generator is 20 + 100 + 50 = 170 kW, and the total reactive power is 70.7 - 28.87 = 41.83 kVAR. The total apparent power is the square root of the sum of the squares of the real and reactive powers, i.e., S = sqrt((170)^2 + (41.83)^2) = 177.3 kVA. The power factor is the ratio of the real power to the apparent power, i.e., PF = P / S = 170 / 177.3 = 0.959, or approximately 0.96 lagging.	
	Q19 (b)	A 2-wire d.c. ring distributor is 300 m long and is fed at 240 V at point A. At point B, 150 m from A, a load of 120 A is taken and at C, 100 m in the opposite 7 direction, a load of 80 A is taken. If the resistance per 100 m of single conductor is 0·03 Ω, find : (i) current in each section of distributor (ii) voltage at points B and C	8 marks
	Ans	(i) Current in each section of the distributor: Let's assume that the current flowing from A to B is I1, and the current flowing from B to C is I2. Since the distributor is a ring circuit, the current returning from C to A is also I1. Using Kirchhoff's first law at point B, we have: I1 = 120 + I2 (1)	

		Using Kirchhoff's first law at point C, we have: I1 = 80 (2) Using Kirchhoff's second law in the loop ABCA, we have: 240 - I1 * 0.03 * 1 - I2 * 0.03 * 0.5 - I1 * 0.03 * 0.5 = 0 Simplifying and substituting for I1 from equation (2), we get: 240 - 0.06 * 80 - I2 * 0.015 - 0.03 * 40 = 0 Solving for I2, we get: I2 = 128 A Substituting I2 in equation (1), we get: I1 = 248 A Therefore, the current in the section from A to B is 248 A, and the current in the section from B to C (and C to A) is 128 A. (ii) Voltage at points B and C: Using Ohm's law, the voltage drop in the section from A to B is: V_AB = I1 * 0.03 * 1 = 7.44 V The voltage at point B is therefore: V_B = 240 - V_AB = 232.56 V Similarly, the voltage drop in the section from B to C (and C to A) is: V_BC = I2 * 0.03 * 0.5 = 1.92 V The voltage at point C is therefore: V_C = V_B - V_BC = 230.64 V Therefore, the voltage at point B is 232.56 V, and the voltage at point C is 230.64 V.	
	Q20 (a)	A 800 metres 2-wire d.c. distributor AB fed from both ends is uniformly loaded at the rate of 1·25 A/metre run. Calculate the voltage at the feeding points A and B if the minimum potential of 220 V occurs at point C at a distance of 450 metres from the end A. Resistance of each conductor is 0·05 Ω/km.	8 marks

	Ans	Given data: Length of 2-wire d.c. distributor, AB = 800 m Load current = 1.25 A/m Resistance of each conductor = 0.05 Ω/km Minimum potential occurs at point C, which is at a distance of 450 m from end A. We can assume that the voltage at point C is equal to the average of the voltages at points A and B. Let the voltage at point A be VA and the voltage at point B be VB. The voltage drop between A and C can be given by: VAC = (1.25 A/m) x (0.05 Ω/km) x (450 m) = 2.8125 V Similarly, the voltage drop between C and B can be given by: VCB = (1.25 A/m) x (0.05 Ω/km) x (350 m) = 1.9375 V Since the voltage at point C is the average of the voltages at points A and B, we can write: VC = (VA + VB)/2 Substituting the values, we get: 220 = (VA + VB)/2 VA + VB = 440 The total voltage drop along the distributor can be given by: VAB = (1.25 A/m) x (0.05 Ω/km) x (800 m) = 40 V The voltage at point A can be given by: VA = VB + VAB Substituting the values, we get: VA = VB + 40 Substituting the value of VA in the equation VA + VB = 440, we get:	

		VB + (VB + 40) = 440 2VB = 400 VB = 200 V Substituting the value of VB in the equation VA = VB + 40, we get: VA = 240 V Therefore, the voltage at point A is 240 V and the voltage at point B is 200 V.	
	Q20 (b)	An industry has a maximum load of 250kW at 0. 707p.f lag, with an annual consumption of 30,000units. The tariff is Rs.60/kVA of maximum demand plus 15 paise per unit. Calculate the following. (i) the flat rate of energy consumption. (ii) annual savings if the p.f is raised to unity	6 marks
	Ans	Given: Maximum load = 250 kW Power factor = 0.707 lag Annual consumption = 30,000 units Tariff = Rs. 60/kVA of maximum demand + 15 paise per unit (i) Flat rate of energy consumption: Let's first calculate the maximum demand in kVA. Maximum demand in kVA = 250 kW / 0.707 = 353.55 kVA The flat rate of energy consumption is given by: Flat rate = Maximum demand in kVA x Tariff rate for maximum demand Flat rate = 353.55 kVA x Rs. 60/kVA Flat rate = Rs. 21,213 The total energy charge is given by: Energy charge = Annual consumption x Tariff rate per unit Energy charge = 30,000 units x Rs. 0.15/unit Energy charge = Rs. 4,500 Total bill = Flat rate + Energy charge = Rs. 21,213 + Rs. 4,500 = Rs. 25,713 (ii) Annual savings if the p.f is raised to unity: New apparent power = 250 kW / 1.0 p.f = 250 kVA New maximum demand in kVA = 250 kVA New bill for maximum demand = 250 kVA x Rs. 60/kVA = Rs. 15,000	

		New energy charge = 30,000 units x Rs. 0.15/unit = Rs. 4,500 New total bill = Rs. 15,000 + Rs. 4,500 = Rs. 19,500 Savings = Old total bill - New total bill = Rs. 25,713 - Rs. 19,500 = Rs. 6,213 Therefore, the annual savings if the power factor is raised to unity is Rs. 6,213.	